CVision

Deliverable 2: Model Architecture and Empirical Evaluation

Track A: Application Development

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Abstract

CVision is an explainable CV (resume) evaluation tool for job seekers: a candidate uploads one PDF and receives an overall quality score with a category breakdown, specific findings, a role-fit score and a skill-gap chart. This report covers Phase 1 (the problem statement, the target audience and the justification for machine learning) and Phase 2 (data acquisition and preprocessing): collection from two public Kaggle datasets (2,484 resume examples in 24 job categories, and a 40,000-posting random sample of 123,849 LinkedIn job postings), cleaning of missing values, duplicates and outliers (leaving 2,439 CVs and 37,162 postings), the feature strategy (normalisation, scaling and encoding), and exploratory data analysis. All code is in the `data_scripts/` folder and the evidence files are in `logs/`. Deliverable 2 adds Phase 3 (a PyTorch dual-branch network over TF-IDF text and 13 structural features, with a 10-trial hyperparameter study) and Phase 4 (stratified 70/15/15 evaluation of five models with Accuracy, Precision, Recall, Macro-F1 and AUC-ROC, plus error analysis and a bias check).

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1 Phase 1: Problem Definition

1.1 Problem Statement

Applying for jobs is now largely automated on the employer's side. Many mid-to-large companies route applications through applicant tracking systems (ATS) that parse, filter and rank CVs before a human reads them.

Candidates, however, get almost no visibility into this process. A rejected applicant rarely receives an explanation, and many never learn whether the problem was weak content, poor formatting, missing skills, or a layout the ATS could not read. This creates several concrete pain points:

- **Silent rejection.** Many applicants never hear back, and those who do often receive a generic rejection. With no feedback, they cannot tell what to change, so they resend the same CV and get the same result.
- **Hidden formatting failures.** Tables, multiple columns, icons and images can confuse ATS parsers, which may scramble or drop entire sections. The candidate sees a clean document, but the system reads something broken.
- **Weak, unquantified content.** Many CVs list duties ("responsible for sales") instead of results ("increased sales by 20%"). Recruiters skim quickly, and bullets without measurable impact are easy to overlook.
- **Unclear skill gaps.** Candidates often do not know which skills their target role demands, so they cannot tell what is missing from their CV or what to learn next.
- **Unaffordable or untrustworthy help.** Professional CV reviews are often expensive, and many free online tools give generic advice or push paid upgrades. Students and fresh graduates in particular may be unable to afford either option.

CVision addresses these problems by turning a single uploaded PDF into a transparent, explainable evaluation: an overall quality score with a category breakdown, specific actionable findings, a role-fit score and a skill-gap chart. The system runs locally on trained models and curated data, without relying on external AI services. CVision gives feedback to help a candidate improve a CV; it does not predict or guarantee hiring outcomes.

1.2 Target Audience

Primary users

- University students and fresh graduates preparing their first CV for internships and entry-level jobs, who typically have no professional feedback available. The concrete core user is a final-year student applying for a first internship or graduate position.
- Early-career professionals changing roles or industries who need to reposition their experience.

Secondary users

- University career-services offices and mentors, who could use the tool to give consistent first-pass feedback to many students at once.
- Bootcamp and training institutes, which could use skill-gap output to guide learners toward in-demand skills.
- Freelancers building profiles for international markets.

1.3 AI Justification

Rule-based code handles measurable properties well (page length, section presence, numbers in bullets, and layout features such as tables or multiple columns detected from the PDF structure), so CVision uses rules for those. The language-heavy parts need machine learning. Role fit will be computed as the cosine similarity between sentence-embedding vectors of the CV and of a job description, and bullet strength by a classifier trained on labelled bullet points. Both will be compared against keyword and regular-expression baselines, so that the benefit of machine learning can be measured rather than assumed.

- **Matching meaning, not keywords.** Rules compare exact strings, so "built predictive models" would never match a "machine learning" requirement. Sentence embeddings place similar meanings close together, so relevance is captured even when the wording differs. This powers the role-fit score.
- **Semantic skill detection.** Whether "Spark," "R," or "Excel" counts as a skill depends on context, and skills are often described rather than named. Comparing CV sentences against a skills vocabulary built from job postings, using embeddings, finds skills even when the wording varies, where keyword lists give false positives or miss variants. This powers the skill-gap chart.
- **Evaluating writing quality.** A regular expression can detect a digit but cannot judge whether a bullet communicates real impact. A classifier trained on labelled bullet points can assess bullet strength more flexibly than a list of banned phrases.
- **Robust section understanding.** CVs have no standard format, and headings vary ("Work History," "Career," "Professional Background"). Semantic models can classify a text block as experience, education or projects even under unfamiliar headings, while heuristics break.
- **Scalability across roles.** Rule-based systems need manual rewriting for each new role or industry. With embeddings, adding a role only requires a description or sample postings, with no new hand-coded rules.
- **Hybrid design by choice.** Rules keep the quality score transparent and auditable, while machine learning supplies semantic understanding. Rules alone are too brittle, and a pure black-box model would be unexplainable, so combining them gives results users can trust and act on.

2 Phase 2: Data Acquisition and Preprocessing

2.1 Data Collection

Two datasets are needed: CV text with a job category (for CV-quality features, section and writing analysis) and job postings (for the role-fit score and the skill vocabulary behind the skill-gap chart). Both were downloaded from Kaggle, and `collect.py` verifies each file and records the evidence in `logs/collection_log.json` (source URL, file size, SHA-256 checksum, row and column counts and a UTC timestamp). With the `--download` option it fetches missing files through the Kaggle command-line tool; otherwise the files are placed in `data/raw/` manually, and the script prints the exact download instructions if a file is missing.

Table 1 summarises both sources. The full dataset pages are listed in the references [1, 2]. The raw files are not stored in the GitHub repository because of their size (the postings file alone is about 517 MB); the checksums in the log allow anyone to confirm that they have downloaded

	Resume Dataset	**LinkedIn Job Postings (2023–2024)**
Kaggle slug	`snehaanbhawal/resume-dataset`	`arshkon/linkedin-job-postings`
Origin	Resume examples scraped from livecareer.com by the dataset author	Job postings scraped from LinkedIn by the dataset author
File used	`Resume.csv`	`postings.csv`
Size	56.27 MB	516.84 MB
Rows × columns	2,484 × 4	123,849 × 31
Main columns	ID, Resume_str, Resume_html, Category	job_id, title, description, skills_desc, formatted_experience_level, formatted_work_type, salary fields, views, applies, ...
Used for	CV-quality features, category analysis	Role-fit scoring, skill vocabulary, skill-gap chart
SHA-256 (first 16 characters)	`816a7cd985a9a41e`	`de33f27c8d1b2260`
Collected	17 Oct 2026 (UTC)	17 Oct 2026 (UTC)
License	CC0: Public Domain	CC BY-SA 4.0

Table 1: Datasets used, as recorded in `collection_log.json`.

identical files. Only the columns needed by the project are read from the postings file, and a random sample of 40,000 postings (seed 42) is used to keep processing fast.

2.2 Data Cleaning

Cleaning is implemented in `clean.py` and covers the three required problems: missing values, duplicates and outliers. Every removed row is written to an audit file (`resumes_audit.csv`, `jobs_audit.csv`) with the reason for its removal, so no record disappears silently, and the totals are stored in `logs/cleaning_report.json`.

Text cleaning (both datasets). HTML tags and entities left over from web scraping are removed, Unicode is normalised (NFKC), decorative symbols are dropped and whitespace is collapsed.

Missing values.

- *CVs:* no raw column contains nulls (Figure 5), but a CV whose text is empty after cleaning, or whose category is missing, cannot be used and is dropped. One CV (a Business-Development CV with 0 words) was removed this way. Category labels are stripped and put in title case.
- *Job postings:* a posting without a title or description is dropped (3 postings in the sample). Missing work type, experience level, company name and location are filled with "Unknown", a missing `remote_allowed` is set to 0, and the very sparse `skills_desc` column is kept with a binary `has_skills_desc` flag. Missing `views` and `applies` are median-imputed and flagged.

Duplicates.

- *CVs:* exact duplicates of the cleaned text are removed (2 found), followed by near-duplicates detected through an MD5 hash of the normalised (lower-case, punctuation-free) text (none found).
- *Job postings:* duplicate job IDs are removed (none found), followed by reposts, defined as the same title, company and normalised description (2,560 found, 6.4% of the sample).

Outliers.

- *Length (both datasets):* documents shorter than a minimum (50 words for CVs, 30 for postings) are not real CVs or postings, and documents above the upper fence $Q_3 + 3.0 \times \text{IQR}$ are treated as extreme outliers. The fences are 1,778 words for CVs and 1,896 words for postings. A factor of 3.0 (instead of the usual 1.5) removes only genuinely extreme documents.
- *Salary:* pay is converted to a yearly figure (hourly × 2,080, weekly × 52, biweekly × 26, monthly × 12), non-positive values and values above 5,000,000 are treated as missing, and the remaining values are winsorised to the 1st and 99th percentiles (92 to 297,157). Because the lower bound is implausibly low, a few erroneous yearly values remain; salary is therefore used only as a weak, optional feature.

Step	CVs	Job postings
Raw rows	2,484	123,849
Random sample (seed 42)	—	40,000
Missing or empty (title/description/text)	1	3
Exact or ID duplicates	2	0
Near-duplicates / reposts	0	2,560
Too short (CV < 50 words, posting < 30 words)	0	207
Too long (above IQR fence)	42	68
Rows kept	**2,439 (98.2%)**	**37,162 (92.9% of sample)**

Table 2: Rows removed at each cleaning step (from `cleaning_report.json`).

Table 2 gives the number of rows removed at every step. The figures in Section 2.4 (missing values, length before and after cleaning) show the effect of these steps.

2.3 Feature Strategy

Feature engineering is implemented in `features.py`. It covers normalisation, scaling and encoding, and all statistics are learned on the training split only to avoid data leakage.

Normalisation. Besides the light cleaning above, an aggressive normalisation is used for text models: lower-casing, removal of all characters except letters, digits, "+" and "#" (so that terms such as C++ and C# survive) and collapsing of whitespace.

CV features (hand-crafted and explainable). Thirteen numeric features are computed for each CV; they feed the transparent part of the quality score (Table 3).

Job features. For postings, the numeric features are the description word count, title word count, length of the skills text, number of vocabulary skills found in the title, description and skills text, yearly salary, views and applies. The 70-skill vocabulary is hand-written (programming, data, cloud, business and soft skills) and is matched by exact keywords; it serves only as a baseline for the skill analysis.

Text encoding. CV text is converted into TF-IDF vectors (unigrams and bigrams, English stop words removed, at least 3 documents per term, sub-linear term frequency, vocabulary limited to 5,000 terms). The vectoriser is fitted on the training CVs only.

Scaling. Numeric features are scaled in two steps. First, $\log(1 + x)$ is applied automatically to every feature whose training-set skewness is greater than 1.0, which compresses long right tails. Second, all numeric features are standardised to zero mean and unit variance with a scaler fitted on the training split. The $\log(1 + x)$ step was applied to `avg_sentence_len`, `metric_number_count`, `percent_count`, `numbers_per_100_words`, `quantified_sentence_ratio` and `action_verb_count` for CVs, and to title length, skills text

Feature	Definition
word_count, sentence_count	Tokens and sentences in the CV
avg_sentence_len, avg_word_len	Mean words per sentence, mean characters per word
lexical_diversity	Unique words divided by words, over the first 300 tokens (length-robust)
metric_number_count	Numbers in the CV that are not years
percent_count	Number of "%" signs
numbers_per_100_words	Non-year numbers per 100 words
quantified_sentence_ratio	Share of sentences containing a non-year number
action_verb_count, action_verbs_per_100_words	Matches against a list of 49 achievement verbs ("led", "increased", ...), raw and per 100 words
section_count	Number of the 7 section types (summary, experience, education, skills, projects, certifications, awards) whose keywords occur in the text
skill_count	Number of skills from a 70-skill vocabulary found in the CV

Table 3: Numeric features computed for each CV.

length, skill count, yearly salary, views and applies for postings. Figure 9 shows the effect on one feature.

Encoding.

- The 24 CV categories are encoded as integers 0–23 (label encoding) and the mapping is stored in `logs/feature_report.json`.
- Job work type (7 values) and experience level (7 values, including "unknown") are one-hot encoded into 14 columns.
- Binary indicators are kept for `remote_allowed`, `has_skills_desc` and `salary_missing`; missing yearly salaries are filled with the median.

Train/validation/test split. The 2,439 CVs are split with stratification by category (random seed 42) into 70% training (1,707), 15% validation (366) and 15% test (366) CVs. Scalers and the TF-IDF vectoriser are fitted on the training split only, so there is no leakage.

2.4 Exploratory Data Analysis

This section interprets the exploratory charts produced by `eda.py`. Two datasets are analysed: the Kaggle *Resume Dataset* (2,484 raw CVs, 24 job categories) and a sample of 40,000 postings drawn from the Kaggle *LinkedIn Job Postings* dataset (123,849 raw postings). Cleaning reduced them to 2,439 CVs and 37,162 postings, respectively.

2.4.1 CV dataset

Figure 1 shows the number of CVs in each of the 24 job categories after cleaning. Twenty-one categories are well balanced, with between 94 and 119 CVs each. Three categories are clearly under-represented: Agriculture (62), Automobile (35) and BPO (21), which gives a largest-to-smallest class ratio of 5.67 (5.45 before cleaning, with 120 CVs in the largest and 22 in the smallest category, so cleaning did not change the class structure). A model trained on this distribution would see far fewer examples of these categories and could perform worse on them, and any per-category statistic for BPO or Automobile rests on very few CVs. We therefore use a stratified 70/15/15 train/validation/test split (1,707 / 366 / 366 CVs), report macro-averaged metrics in addition to accuracy, and apply class weights during training.

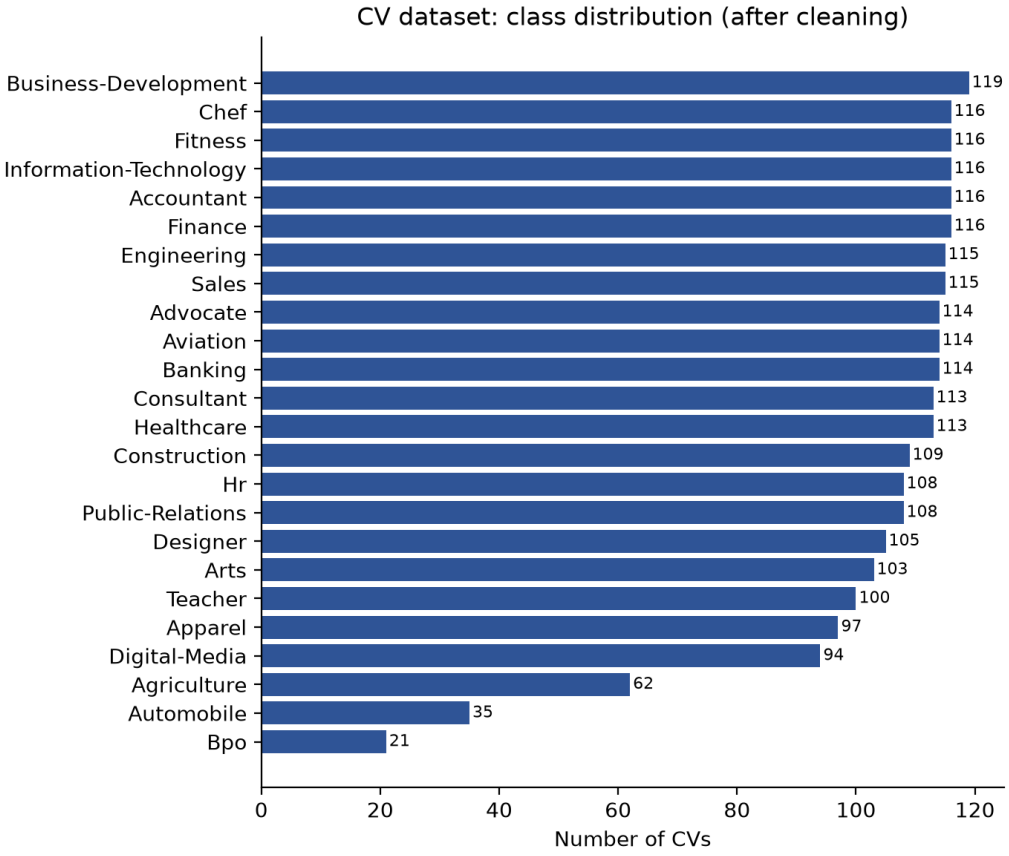


Figure 1: Number of CVs per job category after cleaning.

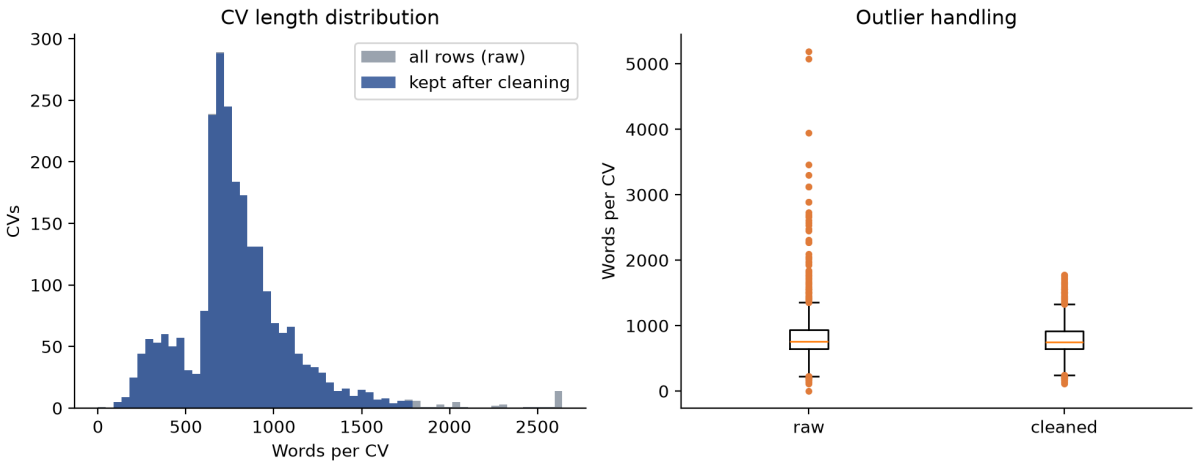


Figure 2: CV length in words, before and after outlier handling (left: histogram, right: box plot). In the histogram, values above the 99.5th percentile of the raw data (≈2,640 words) are clipped into the last bin.

Figure 2 compares CV length before and after cleaning. The cleaned distribution has a median of 751 words, a main peak between roughly 650 and 800 words, a smaller secondary peak around 250–450 words, and a long right tail. In the raw data the longest CVs reach about 5,200 words. The outlier rule removed CVs above the upper fence $Q_3 + 3.0 \times \text{IQR} = 1{,}778$ words (42 CVs, 1.7% of the data); the minimum-length rule (50 words) removed none. The box plot shows that the median and the interquartile range are essentially unchanged by cleaning, so the extreme values were removed without shifting the bulk of the data. We keep the secondary peak of short CVs because it appears to reflect a genuine, concise CV style rather than an error, and CVision's length rule must handle it. Overall, cleaning took the dataset from 2,484 to 2,439 CVs (42 length outliers, 2 exact duplicates and 1 missing record removed), retaining 98.2%.

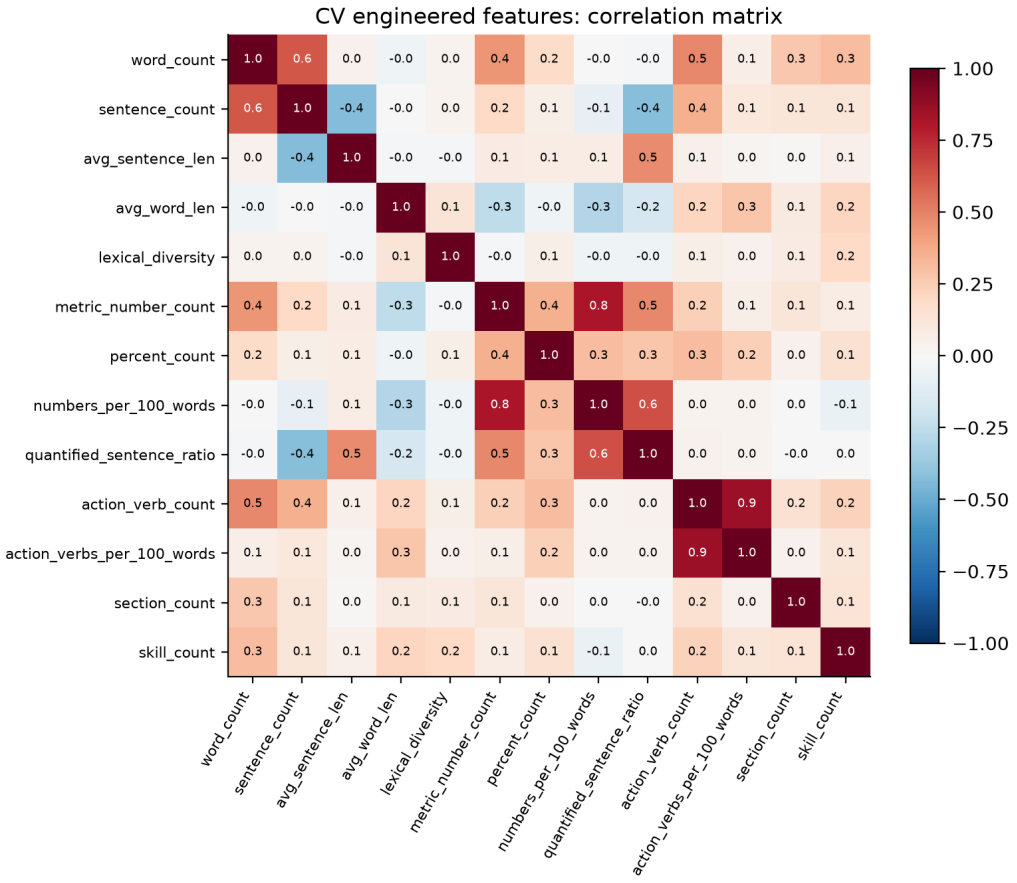


Figure 3: Correlation matrix of the engineered CV features.

Figure 3 shows the pairwise correlations between the 13 engineered CV features. Three groups of strongly related features stand out: the action-verb count and its per-100-word version ($r \approx 0.9$), the metric-number count and numbers per 100 words ($r \approx 0.8$), and word count with sentence count ($r \approx 0.6$). Raw counts also correlate with CV length (for example, action-verb count with word count, $r \approx 0.5$), whereas the per-100-word versions do not (about 0.1 and 0.0). Most other pairs are weak, and `section_count`, `skill_count` and `lexical_diversity` are only weakly related to everything else ($|r| \leq 0.3$), so they carry largely independent information. Note that `section_count` is a keyword-presence heuristic (the number of section keywords such as "experience" or "education" found anywhere in the text), not a true section detector, which explains its weak correlations; real section understanding is left to the models of later phases. The strong pairs are partly redundant. For linear models we will keep the length-independent density features (`numbers_per_100_words`, `action_verbs_per_100_words`) and drop or regularise their raw-count counterparts to avoid multicollinearity; tree-based models are not affected by this.

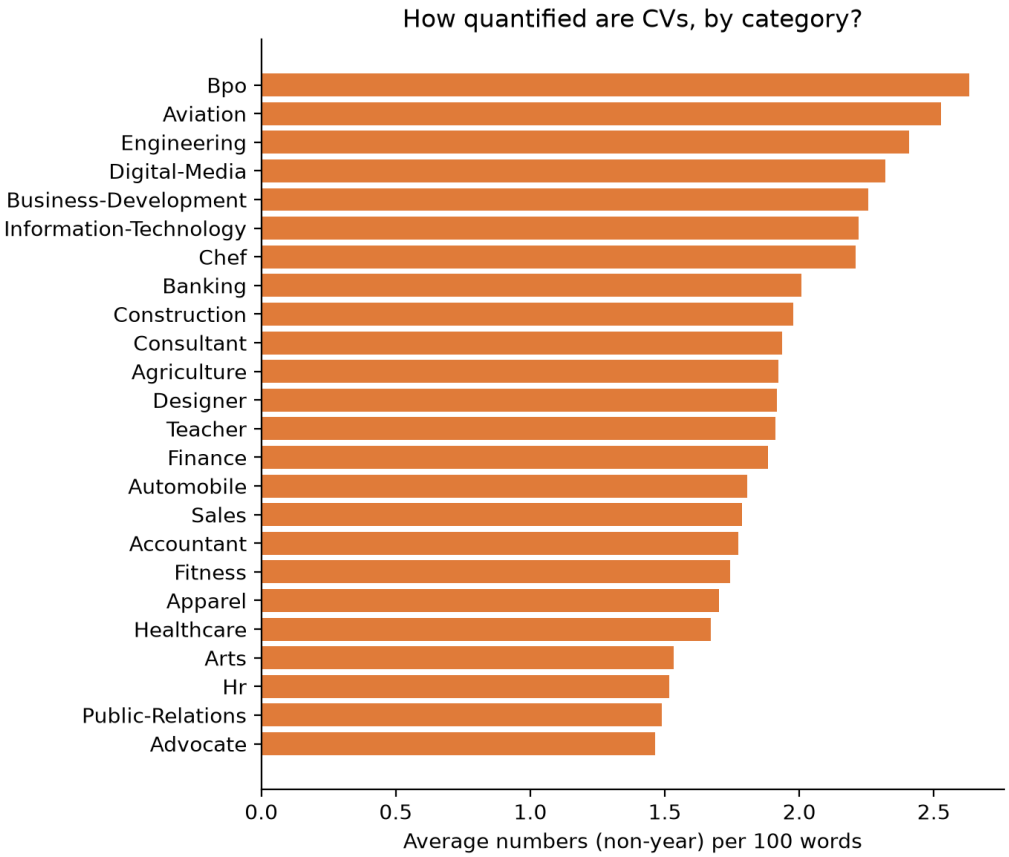


Figure 4: Average number of non-year numbers per 100 words, by CV category.

Figure 4 shows how quantified CVs are in each category, measured as the average number of non-year numbers per 100 words. Values range from about 1.5 (Advocate) to about 2.6 (BPO). Technical and operational categories such as Aviation, Engineering, Digital-Media and Information-Technology sit near the top, while people- and service-oriented categories such as Advocate, Public Relations, HR and Arts sit near the bottom. BPO is the highest, but it has only 21 CVs, so its value is the least reliable. This matters for CVision's "quantified impact" check: a single global threshold would systematically penalise categories in which numbers are naturally less common. The rule should therefore be calibrated per category or expressed relative to the target role.

2.4.2 Job postings dataset

Figure 5 summarises missing data. The CV dataset has no missing values in any of its four columns. The job postings are much less complete: `skills_desc` is empty for 98.0% of postings (121,410 of 123,849), `remote_allowed` for 87.7%, `applies` for 81.2%, both salary bounds for 75.9%, `pay_period` for 70.9%, and `formatted_experience_level` for 23.7%. Company name and views are missing for about 1.4% each, and only 7 postings lack a description. Because `skills_desc` is almost always empty, the skill vocabulary must be derived from the free-text description rather than from this column. Salary is too sparse to serve as a reliable target, so we treat it as an optional feature (winsorised to 92–297,157; the very low lower bound shows that a few implausible yearly values remain, which we did not filter further), and missing experience levels are encoded as an explicit "unknown" category. Missing `views` and `applies` values are median-imputed and flagged, and a missing `remote_allowed` is encoded as 0 (not stated as remote); since these fields are so sparse, they will be treated as weak features. Only postings

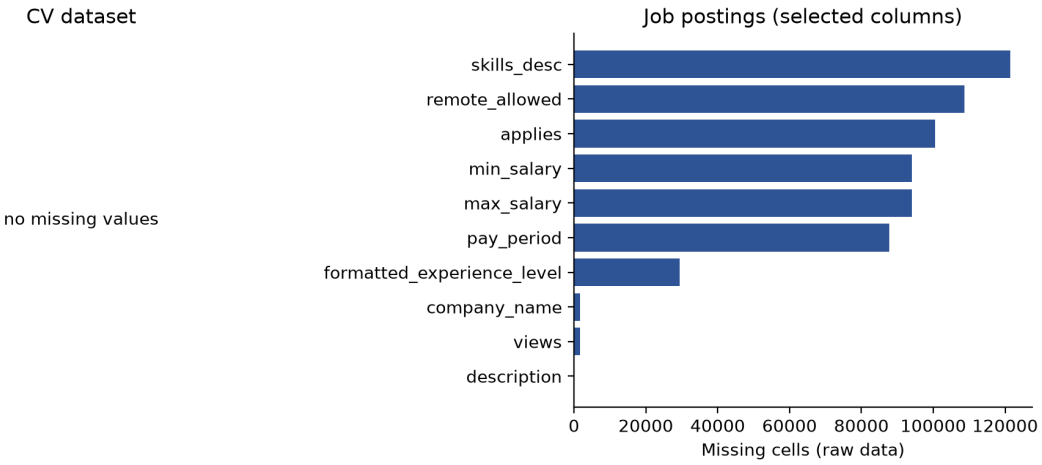


Figure 5: Missing values per column in the raw data.

without a title or description, which cannot be used at all, are dropped (3 in the 40,000-posting sample).

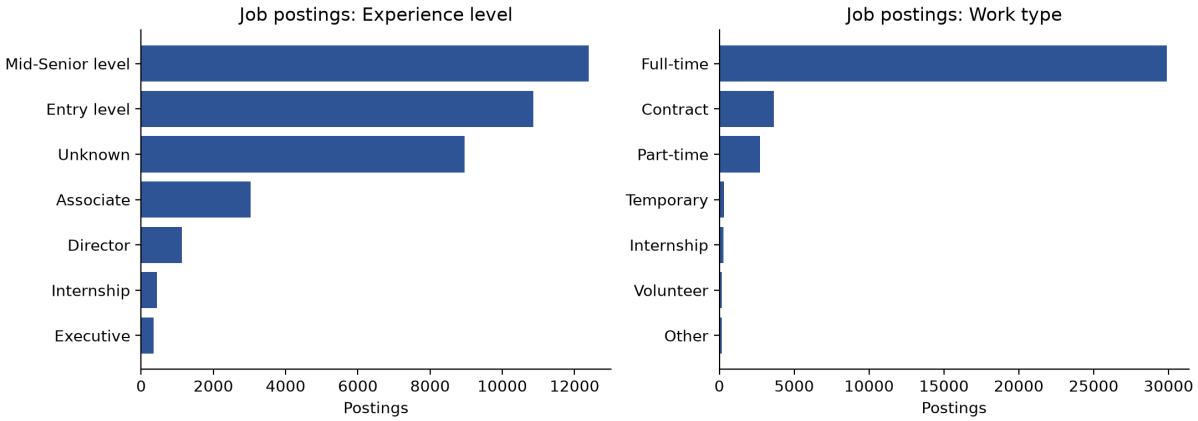


Figure 6: Distribution of experience level (left) and work type (right) in the cleaned postings.

Figure 6 shows the categorical structure of the cleaned postings. Mid-Senior level is the largest experience level (about 12,400 postings), followed by Entry level (about 10,900, roughly 29%) and "Unknown" (about 9,000, which corresponds to the missing values above). Associate, Director, Internship and Executive postings are much rarer, and internships account for only about 1% of the data. Work type is heavily dominated by Full-time (about 29,900 postings, roughly 80%), followed by Contract and Part-time, with Temporary, Internship, Volunteer and Other each under 1%. The sizeable Entry-level share is encouraging for CVision's primary users (students and fresh graduates), but internship-specific postings are scarce, so role-fit results for internships will be less well supported. The very small work-type categories will be grouped or interpreted with caution when used as one-hot features.

Figure 7 shows the length of job descriptions. The cleaned distribution is right-skewed with a median of 482 words and most postings between about 200 and 800 words. Cleaning removed 2,560 duplicate reposts (6.4% of the 40,000-posting sample), 207 postings shorter than 30 words, 68 longer than 1,896 words and 3 with a missing title or description, retaining 37,162 postings (92.9%). The grey-versus-blue comparison shows that duplicate removal reduced the counts across the whole range, while the very short postings, which carry almost no skill information, disappear from the left edge. Postings (median 482 words) are shorter than CVs (median 751 words), and both are long enough that embedding models with a limited input length would

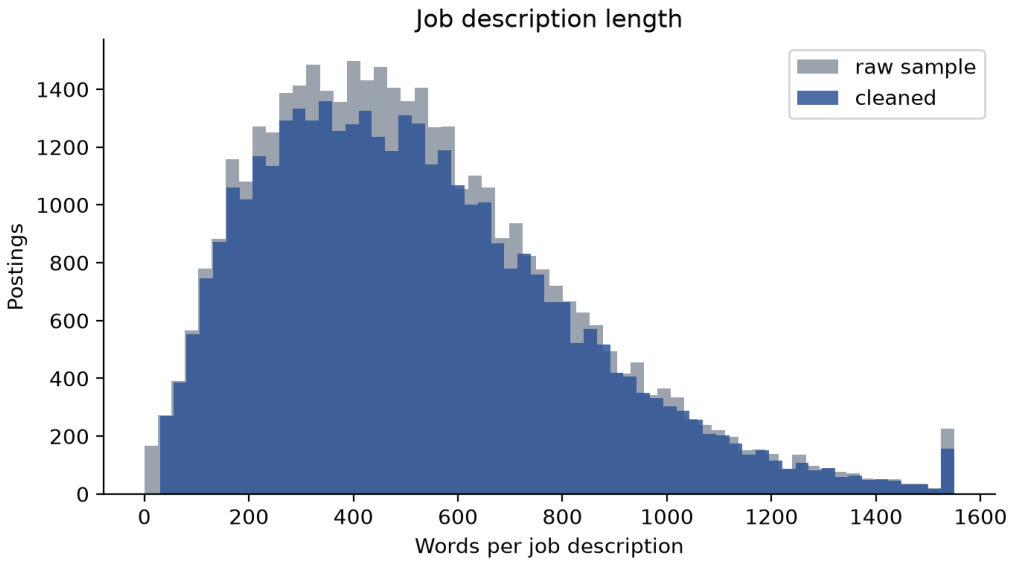


Figure 7: Job description length in words, raw sample versus cleaned data. Values above the 99.5th percentile of the raw sample (≈1,550 words) are clipped into the last bin.

truncate them. Texts will therefore need to be split into chunks, such as by section, before embedding.

2.4.3 Skill demand versus skill supply

Figure 8 previews the skill-gap chart by comparing the share of postings that require a skill with the share of CVs that mention it, using exact keyword matching over a hand-written 70-skill vocabulary; the chart shows the 15 skills with the highest posting share. Communication is the most requested skill (48.7% of postings) and is mentioned about as often on CVs (51.9%). For most other skills CVs mention them far more often than postings require them: sales (about 53% of CVs versus 22%), excel (39% versus 15%), marketing (35% versus 13%), customer service (39% versus 20%) and leadership (41% versus 24%). Only a few skills are demanded more than supplied, notably agile (4.9% versus 2.6%) and teamwork (6.3% versus 4.1%), and among technical skills Python (3.7% versus 1.6%), AWS, Azure, Docker, Kubernetes and machine learning. Six vocabulary skills (TypeScript, deep learning, PyTorch, NLP, Kubernetes and Django) do not appear in any CV. These results should be read with care. The CV dataset spans 24 mostly non-technical occupations and appears to under-represent modern technical profiles, and CVs are longer and more skill-dense than postings, so the dataset-level gap is not a true candidate-level gap. The final CVision chart will instead compare an individual CV with the postings of the chosen target role. This preview also uses exact keyword matching, which cannot recognise paraphrased skills, which motivates embedding-based semantic matching in later phases. We also plan to add more technical CVs to test the technical-role case.

2.4.4 Effect of scaling

Figure 9 shows the effect of the scaling pipeline on `metric_number_count`, the number of non-year numbers in a CV. The raw distribution is strongly right-skewed (skewness 4.53): most CVs contain fewer than about 30 such numbers, while a few exceed 100 and the largest reaches roughly 270. Applying $\log(1 + x)$ followed by z-score standardisation compresses this tail and gives a far more symmetric distribution (skewness -0.78) with zero mean and unit variance, spanning roughly -2.9 to 3.6. The isolated bars on the left are an artefact of integer

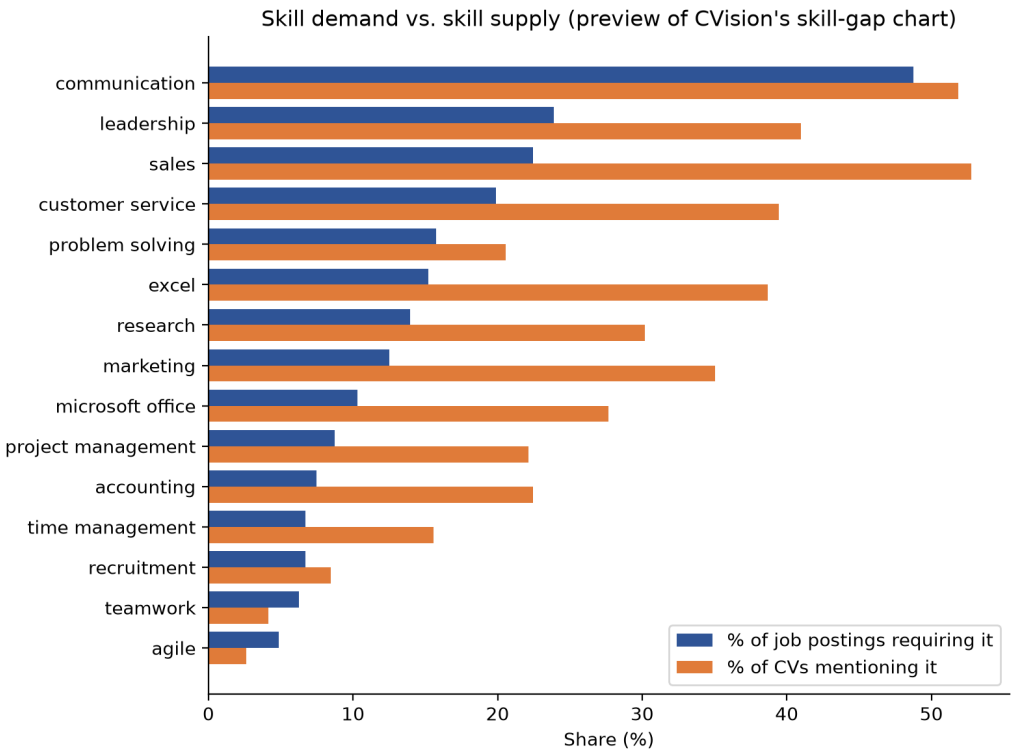


Figure 8: Share of job postings requiring each skill versus share of CVs mentioning it (preview of CVision's skill-gap chart).

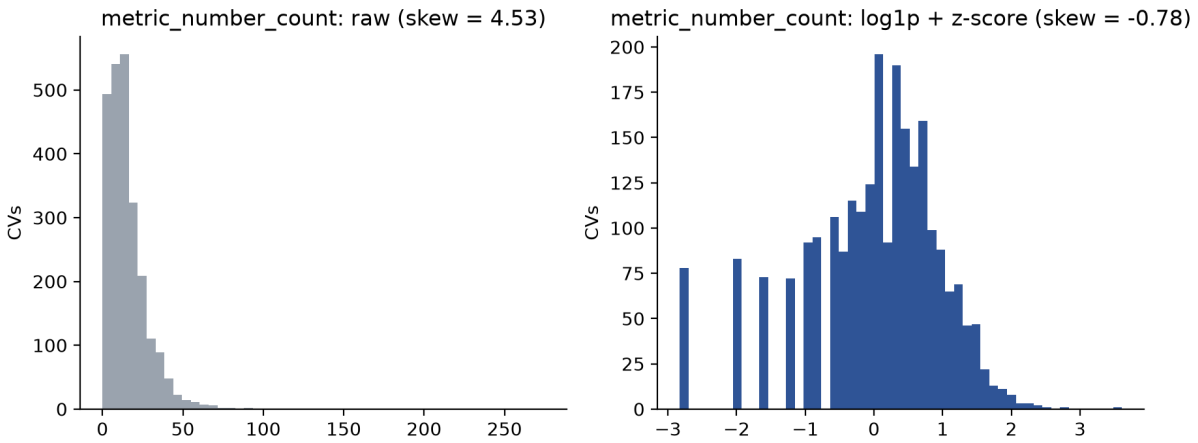


Figure 9: Distribution of `metric_number_count` before (left) and after (right) the $\log(1+x)$ transform and standardisation.

counts, because $\log(1 + x)$ of the smallest values (0, 1, 2, ...) is well separated; the leftmost bar most likely corresponds to CVs containing no numbers at all. The transform slightly over-corrects, leaving a mild left skew, but this is a large improvement for distance-based and regularised models, which would otherwise be dominated by a few extreme CVs. The pipeline applies $\log(1 + x)$ automatically, and only to features whose training-set skewness exceeds 1.0: `avg_sentence_len`, `metric_number_count`, `percent_count`, `numbers_per_100_words`, `quantified_sentence_ratio` and `action_verb_count` for CVs, and title length, skills text length, skill count, yearly salary, views and applies for postings. Less skewed features such as `word_count` (skewness 0.50) are only standardised. Both the skewness test and the scaler statistics are computed on the training split only, so no information from the test set leaks into the features.

3 Phase 3: Model Architecture and Implementation

3.1 Framework and Design

The model is implemented in PyTorch (`src/deep_model.py`); the baselines use Scikit-Learn. The task is 24-class classification of a CV into its job category, using two inputs: the 5,000-dimensional TF-IDF vector x_t and the 13 scaled structural features x_s from Section 2.3. The network has two branches that are fused:

$$
\begin{aligned}
h_t &= \mathrm{Drop}_{0.20}(\mathrm{ReLU}(\mathrm{BN}(W_t x_t + b_t))) \in \mathbb{R}^{256} \\
h_s &= \mathrm{Drop}_{0.10}(\mathrm{ReLU}(\mathrm{BN}(W_s x_s + b_s))) \in \mathbb{R}^{32} \\
h_f &= \mathrm{Drop}_{0.15}(\mathrm{ReLU}(\mathrm{BN}(W_f [h_t; h_s] + b_f))) \in \mathbb{R}^{128} \\
z &= W_o h_f + b_o \in \mathbb{R}^{24}, \qquad \hat{p} = \mathrm{softmax}(z)
\end{aligned}
$$

Figure 10 shows the architecture. The text branch holds most of the parameters (1,280,768 of 1,321,624) because of the wide TF-IDF input. Hidden layers use Kaiming-normal initialisation (suited to ReLU) and the output layer uses Xavier-uniform initialisation. Batch normalisation stabilises training on this small dataset, and the dropout rates are scaled from one base rate: the structural branch uses half of it (capped at 0.20) so that its few dense signals are not erased, and the fusion layer uses 0.75 of it.

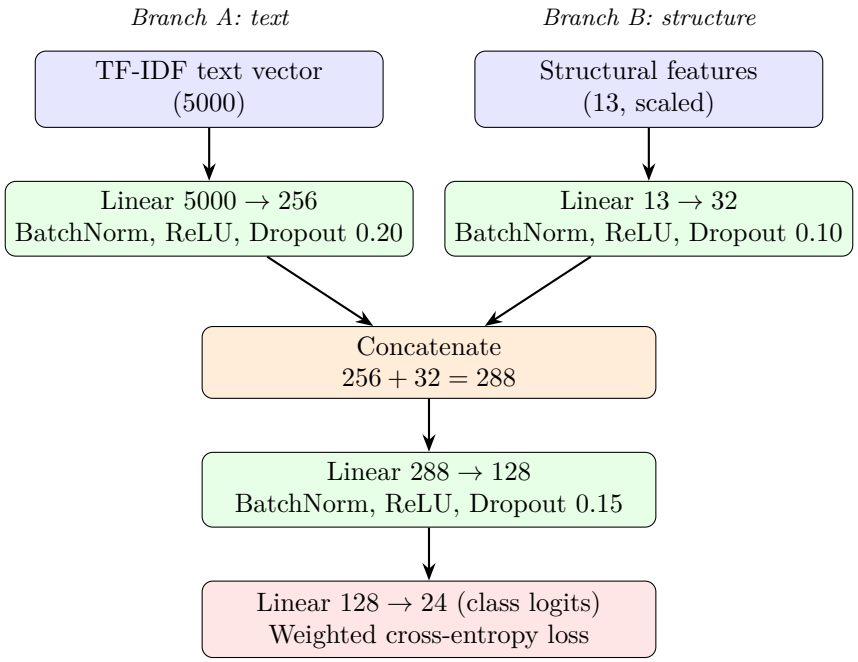


Figure 10: Dual-branch network (1,321,624 trainable parameters). Dropout rates are those of the selected configuration (base rate 0.2).

Ablation. A second model, the text-only ablation, removes the structural branch (fusion input 256 instead of 288). Comparing the two measures whether the 13 engineered features add anything beyond the text.

3.2 Loss Function and Optimiser

Training minimises a class-weighted cross-entropy,

$$\mathcal{L} = -\frac{\sum_i w_{y_i} \log \hat{p}_{i,y_i}}{\sum_i w_{y_i}},$$

where $w_c = \dfrac{N}{K\, n_c}$ is the inverse-frequency weight of class c, $N = 1{,}707$ is the number of training CVs, $K = 24$ the number of classes and n_c the number of training CVs in class c. A perfectly balanced class gets weight 1, and rare categories such as BPO, Automobile and Agriculture get the largest weights, so they are not ignored (Section 2.4). The weights are computed on the training split only.

The optimiser is AdamW (Adam with decoupled weight decay) with learning rate 10^{-3}, weight decay 10^{-4} and batch size 32. Three further mechanisms stabilise training on this small dataset:

- *Learning-rate scheduling:* a `ReduceLROnPlateau` scheduler halves the learning rate when the validation loss has not improved for 3 epochs.
- *Gradient clipping:* the gradient norm is clipped to 1.0.
- *Early stopping and checkpointing:* training runs for at most 25 epochs and stops after 8 epochs without improvement in validation loss. The weights from the epoch with the lowest validation loss are kept and used for the final test evaluation.

All runs use random seed 42. The test set is evaluated only once, after training has finished and the checkpoint has been chosen.

Table 4: Hyperparameter Exploration Grid and Validation Performance

Configuration	Learning Rate	Batch Size	Dropout	Weight Decay	Epochs (Best / Stop)	Val M
Baseline	1e-03	32	0.4000	1e-04	5 / 13	0.
High LR (1e-2)	1e-02	32	0.4000	1e-04	2 / 10	0.
Low LR (1e-4)	1e-04	32	0.4000	1e-04	25 / 25	0.
Small Batch (16)	1e-03	16	0.4000	1e-04	4 / 12	0.
Large Batch (64)	1e-03	64	0.4000	1e-04	8 / 16	0.
Low Dropout (0.2)	1e-03	32	0.2000	1e-04	3 / 11	0.
High Dropout (0.6)	1e-03	32	0.6000	1e-04	8 / 16	0.
No Weight Decay	1e-03	32	0.4000	0	5 / 13	0.
Strong Weight Decay (1e-2)	1e-03	32	0.4000	1e-02	5 / 13	0.
Compact Text (128)	1e-03	32	0.4000	1e-04	6 / 14	0.

3.3 Hyperparameter Tuning

Ten configurations were evaluated on the validation split, varying one factor at a time from the baseline (learning rate 10^{-3}, batch size 32, dropout 0.4, weight decay 10^{-4}, text width 256). Table 4 lists all of them with the epoch of the best (lowest) validation loss, the epoch at which early stopping fired, and the validation Macro-F1 at the selected checkpoint, which was the criterion for choosing between configurations.

Justification of the final selection.

- *Dropout 0.2* gave the best validation Macro-F1 (0.6333, against 0.5968 for the 0.4 baseline), so it was selected. With the 0.2 base rate the effective rates are 0.20, 0.10 and 0.15 in the three layers.
- *Learning rate* 10^{-3} was kept: 10^{-2} was clearly worse (0.5534, unstable), and 10^{-4} had not converged after the full 25 epochs (best epoch 25 of 25).
- *Batch size 32 and weight decay* 10^{-4} were kept. Batch sizes 16 and 64 scored higher than the baseline (0.6186, 0.6184), but both were tested with dropout 0.4, so they were not combined with the winning dropout. Removing weight decay or raising it to 10^{-2} lowered the score.
- *Text width 128* did not help (0.5883), so 256 was kept.

This was a one-factor-at-a-time search, so interactions between settings were not explored. Each configuration was run once on a 366-CV validation set, and differences of a few hundredths in Macro-F1 may be within run-to-run noise; the choice of 0.2 is the best observed, not a proven optimum.

4 Phase 4: Training and Empirical Evaluation

4.1 Validation Strategy

The 2,439 CVs are split once, with stratification by category and random seed 42, into 1,707 training, 366 validation and 366 test CVs (70/15/15). The training split fits the model, scalers and TF-IDF vocabulary; the validation split drives early stopping, checkpoint selection and the hyperparameter study; the test split is used once, for the final results. Stratification matters here because the rarest category (BPO) has only 21 CVs in total, and a random split could leave it almost absent from one split. Even so, some test classes are very small (as few as 2 CVs for

Table 5: Empirical Model Comparison on the 15% Test Split

Model Architecture	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	OvR AUC-ROC
Random Forest	0.7377	0.7281	0.7189	0.6940	0.7149	0.9721
Text Only Ablation	0.6311	0.6241	0.6227	0.5946	0.6166	0.9472
Logistic Regression	0.6284	0.5870	0.6214	0.5856	0.6049	0.9501
Dual Branch Net	0.6175	0.5767	0.5925	0.5737	0.6104	0.9501
Dummy Baseline	0.0464	0.0019	0.0417	0.0037	0.0041	0.5000

Automobile and 4 for BPO), which limits how much any per-class result can be trusted.

4.2 Evaluation Metrics and Their Justification

The categories are imbalanced (largest-to-smallest ratio 5.67), so accuracy alone is misleading: a model can score well by doing well on the large classes while failing on the small ones. We therefore report:

- **Accuracy**, as the familiar overall figure, but not as the main criterion.
- **Macro-Precision, Macro-Recall and Macro-F1**, which average the per-class scores with equal weight, so BPO counts as much as Information Technology. Macro-F1 is our *primary* metric because it exposes failure on rare classes and balances false alarms (precision) against misses (recall).
- **Weighted-F1**, which weights each class by its test frequency and shows how the model performs on a typical CV.
- **One-vs-rest AUC-ROC (macro-averaged)**, which measures how well the predicted probabilities rank the correct class above the others, independently of any decision threshold and of class frequency.

A chance-level Dummy baseline is included as a floor: it reaches 4.64% accuracy, a Macro-F1 of 0.0037 and an AUC-ROC of exactly 0.5, which shows what "no learning" looks like on these metrics.

4.3 Training Behaviour

Figure 11 shows the learning curves of the Dual Branch Net and of the text-only ablation. For the Dual Branch Net, the validation loss is lowest at epoch 3 (1.437), after which it drifts up to about 1.52, and early stopping fires at epoch 11. Meanwhile the training accuracy reaches 100% by epoch 6 and the training loss falls to 0.005, whereas validation accuracy stays near 0.67. The network therefore overfits quickly: it has about 1.3 million parameters and only 1,707 training CVs, and dropout and weight decay slow this down but do not prevent it. Keeping the epoch-3 checkpoint limits the damage. That checkpoint reaches a validation accuracy of 0.678, but a test accuracy of 0.617, a gap that is partly the optimism of selecting the best epoch on the same validation set.

4.4 Model Comparison on the Test Set

Table 5 compares the Dual Branch Net with the text-only ablation and with three baselines (Logistic Regression, Random Forest and the Dummy classifier), all evaluated on the same 366 test CVs.

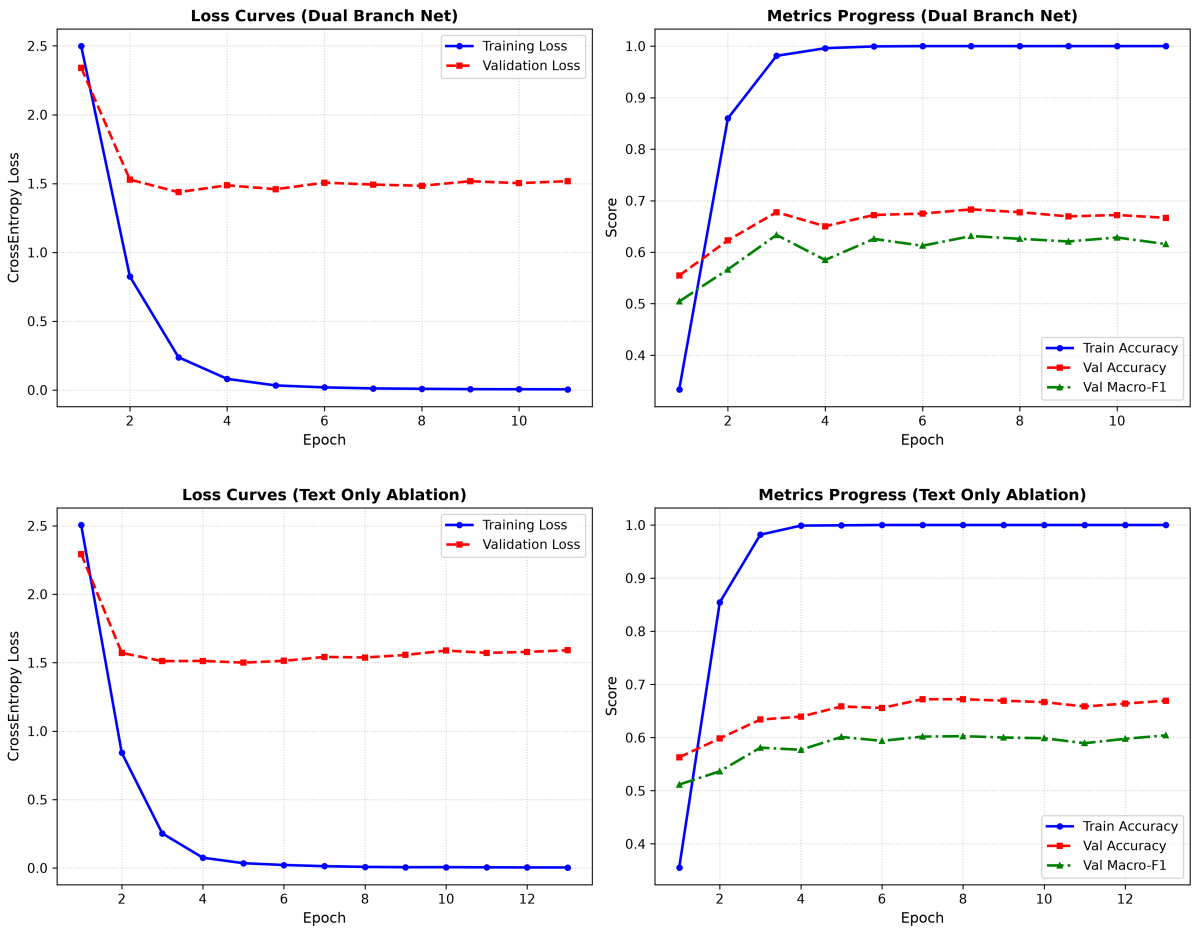


Figure 11: Loss and accuracy/Macro-F1 curves per epoch. Top: Dual Branch Net. Bottom: text-only ablation.

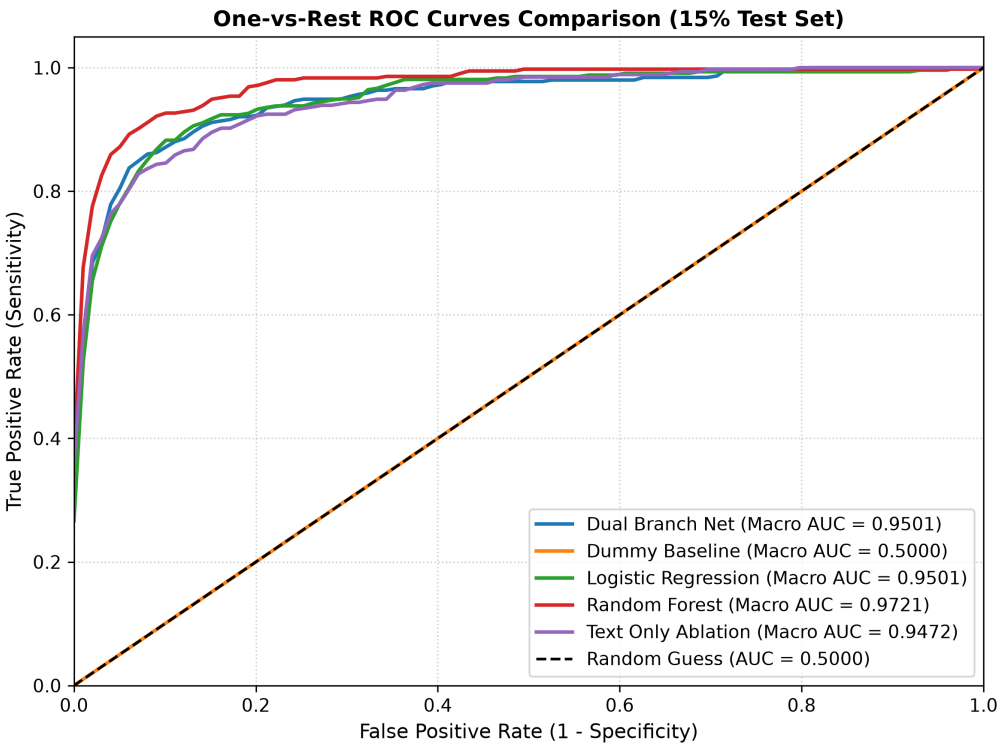


Figure 12: One-vs-rest ROC curves on the test split.

Findings.

- *All learned models are far above the Dummy baseline* (Macro-F1 0.57–0.69 against 0.004), so the TF-IDF text carries a strong signal about the job category.
- *Random Forest is the best model* on every metric (accuracy 0.738, Macro-F1 0.694, AUC-ROC 0.972). The neural network does not beat it, and we report this as it is. A plausible explanation is the small training set: tree ensembles are robust on 1,707 samples with sparse TF-IDF features, while the network overfits (previous section). We did not test this explanation with further experiments.
- *The Dual Branch Net does not beat Logistic Regression* (Macro-F1 0.574 against 0.586; identical AUC-ROC of 0.950). The extra non-linear capacity brings no gain at this data size.
- *The structural features did not help.* The text-only ablation scores slightly higher than the full dual-branch model on accuracy, Macro-F1 and Weighted-F1 (0.631, 0.595, 0.617 against 0.618, 0.574, 0.610), and slightly lower on AUC-ROC (0.947 against 0.950). This is plausible: the 13 features describe CV *quality* (length, numbers, action verbs) rather than *topic*, so they add little for deciding the job category. They remain useful for the transparent quality score of the final application.
- *Caution on small differences.* The test set has 366 CVs, so one CV is 0.27 percentage points of accuracy, and every model was trained with a single seed. Gaps of one or two points between the neural models and Logistic Regression are within the range that noise could explain; only the Random Forest lead and the gap to the Dummy baseline are clearly larger.

4.5 Error Analysis

Figure 13 shows the confusion matrix of the test predictions, and Table 6 lists the best and worst categories of the Dual Branch Net.

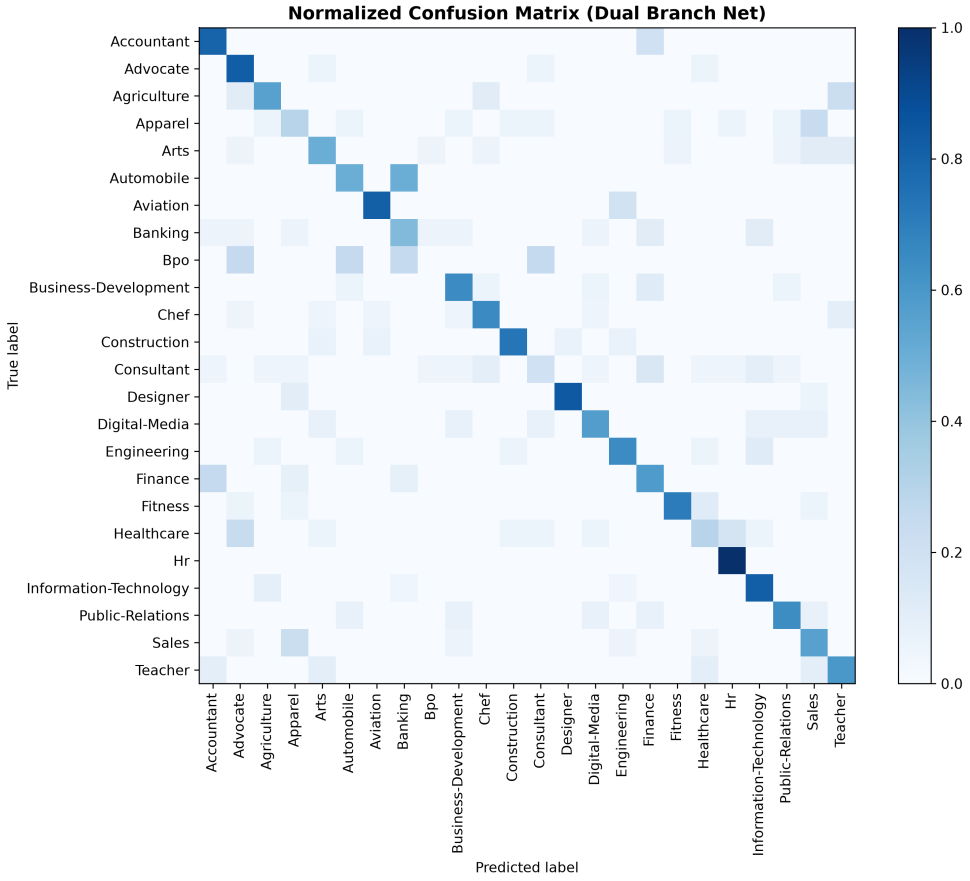


Figure 13: Confusion matrix on the test split.

Category	Precision	Recall	F1	Test CVs
Designer	0.941	0.842	0.889	19
HR	0.783	1.000	0.878	18
Aviation	0.867	0.813	0.839	16
Healthcare	0.417	0.294	0.345	17
Apparel	0.333	0.294	0.313	17
Consultant	0.444	0.200	0.276	20
Automobile	0.167	0.500	0.250	2
BPO	0.000	0.000	0.000	4

Table 6: Best three and worst five categories of the Dual Branch Net (from `per_class_dual_branch_net.csv`).

Where the model fails.

- *Rare classes.* BPO (4 test CVs) is never predicted correctly (F1 0.00), and Automobile (2 test CVs) reaches only 0.25. These are the categories flagged as under-represented in the EDA, and the inverse-frequency weights were not enough to fix them.
- *Overlapping professions.* The most frequent confusions are between neighbouring occupations: Sales and Apparel (4 CVs each way), Finance and Accountant (3 each way),

Table 7: Subgroup Bias Audit Across Category Representation, Document Length, and Gender Proxy

Subgroup Audit Dimension	Test Support	Accuracy / F1	Disparity Finding
Majority Categories (Top 5)	18–22	0.6265	Baseline Reference
Minority Categories (Bottom 5)	2–12	0.3577	Delta-F1: 0.2688
Length: Short (<669w)	122	0.6230	Confidence: 0.579
Length: Medium (669-842w)	122	0.6230	Confidence: 0.601
Length: Long (>842w)	122	0.6066	Confidence: 0.626
Gender/Title Perturbation	2 feature-changed (366 total)	1.0000 consistency	Flip Rate: 0.00 percent

Consultant predicted as Finance (3) or Information Technology (2), Healthcare predicted as Advocate (4) or HR (3), Aviation predicted as Engineering (3), and Designer predicted as Apparel (2). Such CVs share vocabulary, and a CV with a mixed profile has only one label, so some of these errors reflect label ambiguity and not only model weakness.

- *Not only a data-size problem.* Consultant has 20 test CVs, one of the largest classes, yet an F1 of only 0.28, because consulting CVs describe work in finance, IT and other domains. Class size alone does not explain the errors.
- *Distinctive classes work well.* Designer, HR and Aviation have the clearest vocabulary and reach F1 between 0.84 and 0.89. HR has a recall of 1.00 but a precision of 0.78, because it also absorbs some Healthcare CVs.

The high AUC-ROC (about 0.95) next to a 62% accuracy suggests that the correct category is often ranked highly even when it is not first. For CVision this points to showing the top few candidate categories, and not a single hard label, in the final application. We did not compute top-k accuracy in this deliverable.

4.6 Bias Check

The dataset has no demographic attributes, so we cannot audit subgroups such as age or ethnicity directly. We run three audits on the Dual Branch Net instead (Table 7, Figure 14).

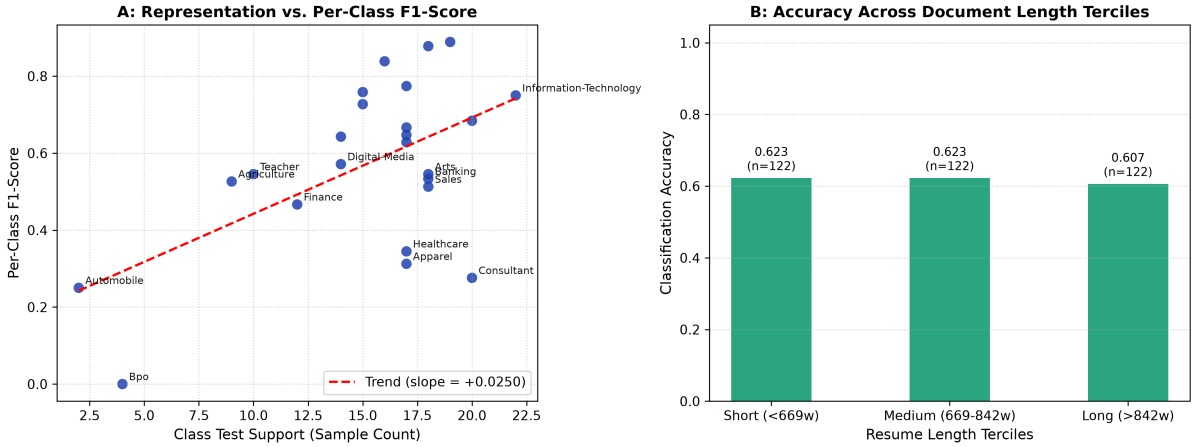


Figure 14: Bias check of the Dual Branch Net.

1. **Class representation (clear bias).** The five best-represented categories reach a mean F1 of 0.627, the five least represented only 0.358, a gap of 0.269. Across the 24 classes, F1 is positively correlated with class size (Pearson $r = 0.555$, $p = 0.0049$). The model is

therefore less reliable for under-represented occupations, despite the class-weighted loss. This matters for CVision: a CV from a rare category would receive a less trustworthy classification.

2. **Document length (small effect).** Accuracy is 0.623 for short CVs (<669 words), 0.623 for medium CVs (669–842) and 0.607 for long CVs (>842), with 122 CVs in each group. The long group differs by about two CVs, which is too small to call a bias. The mean prediction confidence rises with length (0.579, 0.601, 0.626) while accuracy does not, so the model is slightly more over-confident on long CVs.
3. **Gender proxy (inconclusive).** We swapped gendered words and pronouns in the test CVs and counted changed predictions. The flip rate is 0.00%, but only 2 of the 366 CVs contained any such feature, so this is weak evidence and not proof of fairness. It also tests only explicit gendered words, not indirect proxies such as names or hobbies.

Limits and mitigation. The CVs come from one website with a template style, and most of the 24 occupations are non-technical, so these results may not transfer to other CV populations. Reasonable next steps are collecting more CVs for BPO, Automobile and Agriculture, oversampling them, and testing fairness on a dataset that has demographic labels.

5 Team Contribution Table

Table 8 maps the Git commit authors to the project modules. Each student appears under two or more Git identities (the account name and the name configured on their machine); `git shortlog -sn -all` shows them. The table is reproducible from `git log --name-only`.

Member	Git author names	Modules and work (from the commit history)
Aqsa Ehtesham (24L-3017)	`AqsaEhtesham`, `Aqsa Ehtesham Khan` (pull-request merges)	`src/dataset.py` (stratified 70/15/15 split, class weights), `src/baseline_models.py` (Dummy, Logistic Regression, Random Forest), `src/evaluate.py` (metrics, ROC curves, confusion matrices, per-class reports); Deliverable 1 EDA figures, `eda.py` fixes and the first LaTeX report.
M. Shehryar Waheed (24L-3023)	`Muhammad Shehryar Waheed`	`src/deep_model.py` (dual-branch network), `src/train.py` (AdamW, weighted loss, scheduler, early stopping), `src/tune.py` (10-configuration study), `src/bias_check.py` (Phase 4 bias audit), `src/quality_score.py`; the final comparison run with all five models; Deliverable 1 pipeline (`data_scripts/`) and repository set-up.

Table 8: Contribution table: Git commit authors and project modules.

References

[1] Kaggle user snehaanbhawal. *Resume Dataset*. Kaggle. `https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset` (accessed 17 October 2026).

[2] Kaggle user arshkon. *LinkedIn Job Postings (2023–2024)*. Kaggle. `https://www.kaggle.com/datasets/arshkon/linkedin-job-postings` (accessed 17 October 2026).